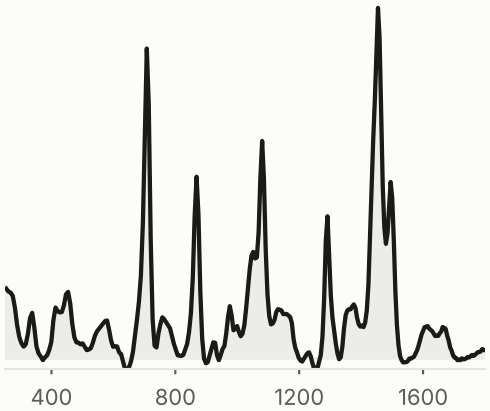
A computed Raman library does not replace measured reference spectra

Eleven authentic chemical warfare agents at 1064 nm · same queries, same pipeline, same metric · only the library differs

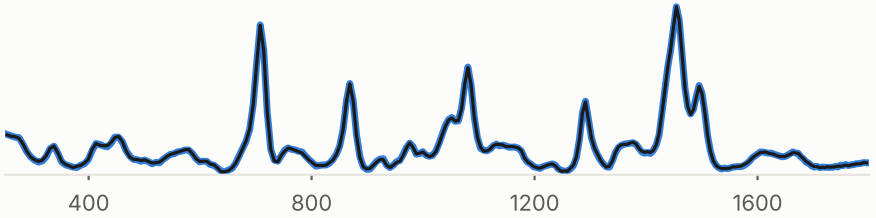
QUERY
one acquisition of an authentic agent



Raman shift (cm^{-1})

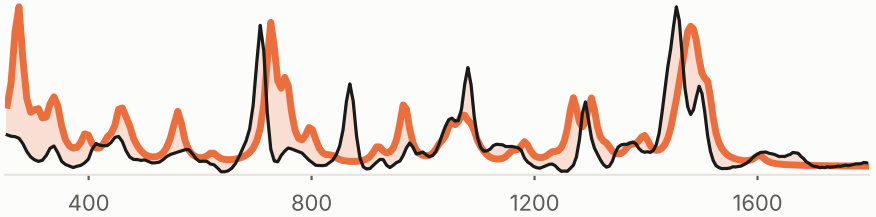
the thin black trace is this query, repeated on both panels;
the thick coloured trace is the library entry it is compared with

measured reference library — **cosine 1.00**



30/30 acquisitions identified correctly

computed library — **cosine 0.65**



13/30 acquisitions and no repair closes it

shaded area is the disagreement · A-232 shown; the set mean is 0.67 against 0.98 for the measured library

Computed library: B3LYP/6-311++G(d,p), 54 members, calibrated on band positions alone with no parameter fitted to identification.
The best of twelve similarity metrics reaches 16 of 30; two corrected structures, an affine wavenumber axis, a ±30 cm^{-1} offset scan and symmetric preprocessing all leave it at 13.